MSG No. 123.346 $P4'/m'mm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `1a`, site symmetry: `4'/m'mm'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `1b`, site symmetry: `4'/m'mm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `1c`, site symmetry: `4'/m'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `1d`, site symmetry: `4'/m'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,5,6,9,10,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,5,6,9,10,15,16]`

Table 7: Wyckoff site: `2g`, site symmetry: `4'mm'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,7,8,9,10,15,16]`
2	$[0, 0, -z]$	`[3,4,5,6,11,12,13,14]`

Table 8: Wyckoff site: 2h, site symmetry: 4'mm'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[1,2,7,8,9,10,15,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[3,4,5,6,11,12,13,14]

Table 9: Wyckoff site: 4i, site symmetry: 2mm.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,2,7,8]
2	$\left[\frac{1}{2}, 0, -z\right]$	[3,4,5,6]
3	$\left[\frac{1}{2}, 0, z\right]$	[9,10,15,16]
4	$\left[0, \frac{1}{2}, -z\right]$	[11,12,13,14]

Table 10: Wyckoff site: 4j, site symmetry: m'.2m'

No.	position	mapping
1	$[x, x, 0]$	[1,3,14,16]
2	$[-x, -x, 0]$	[2,4,13,15]
3	$[x, -x, 0]$	[5,8,10,11]
4	$[-x, x, 0]$	[6,7,9,12]

Table 11: Wyckoff site: 4k, site symmetry: m'.2m'

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	[1,3,14,16]
2	$\left[-x, -x, \frac{1}{2}\right]$	[2,4,13,15]
3	$\left[x, -x, \frac{1}{2}\right]$	[5,8,10,11]
4	$\left[-x, x, \frac{1}{2}\right]$	[6,7,9,12]

Table 12: Wyckoff site: 4l, site symmetry: m'2'm.

No.	position	mapping
1	$[x, 0, 0]$	[1,8,11,14]
2	$[-x, 0, 0]$	[2,7,12,13]
3	$[0, x, 0]$	[3,6,9,16]
4	$[0, -x, 0]$	[4,5,10,15]

Table 13: Wyckoff site: 4m, site symmetry: m'2'm.

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,8,11,14]
2	$\left[-x, 0, \frac{1}{2}\right]$	[2,7,12,13]
3	$\left[0, x, \frac{1}{2}\right]$	[3,6,9,16]

continued ...

Table 13

No.	position	mapping
4	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[4,5,10,15]`

Table 14: Wyckoff site: `4n`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,8,11,14]`
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2},\ x,\ 0\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[4,5,10,15]`

Table 15: Wyckoff site: `4o`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,8,11,14]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[4,5,10,15]`

Table 16: Wyckoff site: `8p`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$[-x,\ -y,\ 0]$	`[2,13]`
3	$[y,\ x,\ 0]$	`[3,16]`
4	$[-y,\ -x,\ 0]$	`[4,15]`
5	$[y,\ -x,\ 0]$	`[5,10]`
6	$[-y,\ x,\ 0]$	`[6,9]`
7	$[-x,\ y,\ 0]$	`[7,12]`
8	$[x,\ -y,\ 0]$	`[8,11]`

Table 17: Wyckoff site: `8q`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,14]`
2	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[2,13]`
3	$\left[y,\ x,\ \frac{1}{2}\right]$	`[3,16]`
4	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[4,15]`
5	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[5,10]`
6	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[6,9]`
7	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[7,12]`
8	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[8,11]`

Table 18: Wyckoff site: 8r, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$[-x,\ -x,\ z]$	[2,15]
3	$[x,\ x,\ -z]$	[3,14]
4	$[-x,\ -x,\ -z]$	[4,13]
5	$[x,\ -x,\ -z]$	[5,11]
6	$[-x,\ x,\ -z]$	[6,12]
7	$[-x,\ x,\ z]$	[7,9]
8	$[x,\ -x,\ z]$	[8,10]

Table 19: Wyckoff site: 8s, site symmetry: .m.

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,8]
2	$[-x,\ 0,\ z]$	[2,7]
3	$[0,\ x,\ -z]$	[3,6]
4	$[0,\ -x,\ -z]$	[4,5]
5	$[0,\ x,\ z]$	[9,16]
6	$[0,\ -x,\ z]$	[10,15]
7	$[x,\ 0,\ -z]$	[11,14]
8	$[-x,\ 0,\ -z]$	[12,13]

Table 20: Wyckoff site: 8t, site symmetry: .m.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	[1,8]
2	$\left[-x,\ \frac{1}{2},\ z\right]$	[2,7]
3	$\left[\frac{1}{2},\ x,\ -z\right]$	[3,6]
4	$\left[\frac{1}{2},\ -x,\ -z\right]$	[4,5]
5	$\left[\frac{1}{2},\ x,\ z\right]$	[9,16]
6	$\left[\frac{1}{2},\ -x,\ z\right]$	[10,15]
7	$\left[x,\ \frac{1}{2},\ -z\right]$	[11,14]
8	$\left[-x,\ \frac{1}{2},\ -z\right]$	[12,13]

Table 21: Wyckoff site: 16u, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[y,\ x,\ -z]$	[3]
4	$[-y,\ -x,\ -z]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$[-x,\ y,\ z]$	[7]

continued ...

Table 21

No.	position	mapping
8	$[x, -y, z]$	[8]
9	$[-y, x, z]$	[9]
10	$[y, -x, z]$	[10]
11	$[x, -y, -z]$	[11]
12	$[-x, y, -z]$	[12]
13	$[-x, -y, -z]$	[13]
14	$[x, y, -z]$	[14]
15	$[-y, -x, z]$	[15]
16	$[y, x, z]$	[16]